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Near-infrared spectroscopic models for analysis of winter pea (*Pisum sativum* L.) quality constituents

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Abstract

BACKGROUND: Winter pea (*Pisum sativum* L.) grows well in a wide geographic region, both as a forage and cover crop. Understanding the quality constituents of this crop is important for both end uses; however, analysis of quality constituents by conventional wet chemistry methods is laborious, slow and costly. Near infrared reflectance spectroscopy (NIRS) is a precise, accurate, rapid and cheap alternative to using wet chemistry for estimating quality constituents. We developed and validated NIRS calibration models for constituent analysis of this crop.

RESULTS: Of the 11 constituent models developed, nine constituents including moisture, dry-matter, total-nitrogen, crude protein, acid detergent fiber, neutral detergent fiber, AD-lignin, cellulose and non-fibrous carbohydrate had low standard errors and a high coefficient of determination ($R^2 = 0.88 - 0.98$; 1 − VR, which is the coefficient of determination during cross-validation = 0.77 − 0.92) for both calibration and cross-validation, indicating their potential for quantitative predictability. The calibration models for ash ($R^2 = 0.65$; 1 − VR = 0.46) and hemicellulose ($R^2 = 0.75$; 1 − VR = 0.50) also appeared to be adequate for qualitative screening. Predictions of an independent validation set yielded reliable agreement between the NIRS predicted values and the reference values with low standard error of prediction (SEP), low bias, high coefficient of determination (r^2 = 0.82-0.95), high ratios of performance to deviation (RPD = SD/SEP; 2.30 − 3.85) and high ratios of performance to interquartile distance (RPIQ = IQ/SEP; 2.57 − 7.59) for all 11 constituents.

CONCLUSION: Precise, accurate and rapid analysis of winter pea for major forage and cover crop quality constituents can be performed at a low cost using the NIRS calibration models developed.

Keywords: winter pea; calibration; cover crop quality; forage quality; near-infrared spectroscopy

INTRODUCTION

Legume cover crops are recognized for their ability to provide substantial nitrogen (N) to the following crop;[1,2] this helps conventional producers reduce synthetic fertilizer inputs and is a critical source of N fertility for organic producers. Beyond fertility provision, legume cover crops also improve soil quality and can aid in insect and disease suppression.[3–5]

Winter pea (*Pisum sativum* L.) is a legume that has desirable attributes for use as a forage and cover crop in the southeast. Winter pea can produce high biomass in this region, allowing for substantial N delivery to the following cash crop. In North Carolina, winter pea provided 70–91 kg N ha^{-1} in the Coastal Plain and 128–208 kg N ha^{-1} in the Piedmont regions following termination.[2] Other research conducted in the southeast showed that winter pea could produce more dry matter than hairy vetch.[6] Beyond biomass production and N provision, additional benefits of winter pea include easy termination and no hard seed.[7] Pea has a high nutritive concentration: 18–30% protein, 35–50% starch and considerable concentrations of the amino acid lysine, which is deficient in small grains.[8] Pea can be grazed, cut for hay and ensiled.[9,10] Pea silage is more degradable in ruminant animals than barley silage.[11] For these reasons, winter pea is a desirable forage crop. Therefore, there is a great need for analysis of this crop for its forage and cover crop quality constituents, namely moisture, dry-matter (DM), total-nitrogen (TN), crude protein (CP), acid detergent fiber (ADF), neutral detergent fiber (NDF), AD-lignin, cellulose and non-fibrous carbohydrate (NFC). However, the conventional laboratory reference methods for these analyses are laborious, time consuming and costly. By contrast,

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the near infra-red spectroscopic (NIRS) technique, if based on properly developed and validated calibration models, could be an excellent alternative to provide rapid and low cost analyses without sacrificing preciseness and accuracy.

All plant biomass is composed primarily of the three atoms, C, H and O.[12,13] The basis for NIRS analysis is molecular vibrations. An asymmetric molecule, such as CH_3-CH_2-OH or H_2O, vibrates when exposed to NIR light, whereas symmetric molecules, such as Cl_2 or $H_2C{=}CH_2$, are not NIR active. Such NIR active bonds include O−H (as in water, alcohols and phenols), C−H, C−O, C−O−H, N-H and C=C bonds, covering most of the covalent bonds of various organic compounds in the plant biomass,[14] which interact with NIR light and yield quantitative information on the sample's molecular composition accurately and rapidly, keeping the sample intact.[15−17]

The NIRS has been of particular interest for forage analysis with respect to providing various analytical needs for forage-livestock production systems.[18] However, there is no properly developed and validated NIRS calibration model specific to winter pea analysis despite its great potential for use as both a forage and a cover crop. The calibration models for alfalfa, other legumes and a mixture of legume and grass available elsewhere could merely serve qualitative screening tool or may not useful at all for this crop. The present study reports the successful development and validation of NIRS calibration models for the analysis of this crop to meet analytical needs of the farming industries that would utilize this crop as a forage or a cover crop.

MATERIALS AND METHODS

Samples and sample preparation

The samples used to develop and validate NIRS calibration models in the present study were obtained from field research conducted during the 2015/16 and 2016/17 growing seasons in Maryland and North Carolina. From 2015/16, this experiment was conducted at two sites on the USDA-ARS Agricultural Research Center in Beltsville, MD (North: 39.03°N, 76.93°W; South: 39.02°N, 76.94°W); the Central Crops Research Station in Clayton, NC (35.67°N, 78.51°W); and the Caswell Research Farm in Kinston, NC (35.16°N, 77.36°W). From 2016/17, the experiment was conducted at the Central Crops Research Station in Clayton, NC; the Upper Coastal Plain Research Station in Rocky Mount, NC (35.90°N,77.68°W); and the Piedmont Research Station in Salisbury, NC (35.41°N, 80.37°W). The Beltsville North trial was conducted on a Hammonton loamy sand (coarse-loamy, siliceous, semiactive, mesic Aquic Hapludults); the Beltsville South trial was conducted on a Hatboro silt loam (fine-loamy, mixed, active, non-acid, mesic Fluvaquentic Endoaquepts); the Clayton 2015/16 trial was conducted on a Varina loamy sand (fine, kaolinitic, therminc Plinthic Paleudults); the Clayton 2016/17 trial was conducted on a Norfolk loamy sand (fine-loamy, kaolinitic, thermic Typic Kandiudults); the Kinston trial was conducted on a Johns sandy loam (fine-loamy over sandy or sandy-skeletal, siliceous, semiactive, thermic Aquic Hapludults); the Rocky Mount trial was conducted on a Goldsboro fine sandy loam (fine-loamy, siliceous, subactive, thermic Aquic Paleudults); and the Salisbury trial was conducted on a Lloyd clay loam (fine, kaolinitic, thermic Rhodic Kanhanpludults).

The objective of this research was to screen and compare winter pea genotypes for forage and cover crop use in the southeastern USA. Eighteen pea genotypes were screened at each experimental site in a randomized complete block design with three to four replications. Seventeen pea genotypes were obtained from the USDA-ARS winter pea breeding program at Pullman, WA. Eleven of these genotypes were released pea cultivars: cv.s 'CAH-11', 'Chelan', 'Common', 'Fenn', 'Glacier', 'Granger', 'Lynx', 'Melrose', 'Romack', 'Specter' and 'Windham', and six were advanced ARS breeding lines: PS07300136W, PS09300095W, PS10300120W, PS10300121W, PS06300028W and PS0017018W. The winter pea genotype readily available in North Carolina (cv 'Nature's Choice') was also screened. Pea was drilled at 444 600 seeds ha^{-1}.

From winter growth, pea biomass at 0.5 cm above-ground was collected over four sampling events with the initial sampling event occurring when the first flowering pea genotype appeared and then 2, 4 and 6 weeks after the initial sampling event. During the 2015/16 growing season, the actual sampling period in Maryland was from 2 May to 13 June 2016 and that in North Carolina was from 8 April 19 May 2016; the same during the 2016/17 growing season in North Carolina was from 7 April to 22 May 2017. Pea biomass was harvested from two replications per sampling event using a 0.5-m^2 quadrat randomly placed within the plot. Harvested samples were dried at 65 °C to obtain dry weight. Samples were then ground using a Thomas-Willy Laboratory Mill (Arthur H Thomas Company, Philadelphia, PA, USA) and passed through a 1-mm sieve to homogenize biomass. Samples were stored at room temperature prior to NIR spectra analyses followed by wet chemistry analyses on a subset of samples for various quality constituents of interest, as described below. NIR spectra were collected on all samples (n = 945); 507 of these samples were collected from the 2015/16 growing season and 438 samples were collected from the 2016/17 growing season. A sample subset, representing 16% of the total number of samples (n = 152), was analyzed via wet chemistry as described below.

Analyses of samples by laboratory reference methods

Dry matter content of the samples was determined by drying approximately 2 g of sample in a forced-air drying oven at 135 ± 2 °C for 2 h with freely circulating air in accordance with the Association of Official Analytical Chemist (AOAC) method 930.15.[19] Total nitrogen content was determined using 0.2 g of the ground and dried sample by a combustion N analyzer (model 'Rapid N Cube'; Elementar Analysensysteme GmbH, Langenselbold, Germany) following a dry combustion method[20,21] based on the original method described by Dumas.[22] Crude protein content was calculated by multiplying the total nitrogen content by 6.25. Ash content was determined based on ASTM standard D3174-97 for coal and coke.[23]

NDF and ADF analyses were carried out using an Ankom$^{200/220}$ Fiber Analyzer (Ankom Technology, Macedon, NY, USA) using F57 filter bags (Ankom Technology) constructed from chemically inert and heat resistant filter media, capable of being heat sealed closed and able to retain 25-μm particles at the same time as permitting rapid solution penetration.[24,25] The protocols are based on the basic principles of methods 5.1 and 4.1 of the National Forage Testing Association.[26,27] The 'Acid-Detergent Lignin (AD-lignin)' and ash contents in ADF residue were determined in accordance with the method described by Ankom Technology.[28] The fat content was analyzed gravimetrically in accordance with ANKOM Technology Method 2, 01-30-09.[29]

Hemicellulose and cellulose contents were estimated from NDF, ADF, lignin and ash as follows:

$$\%\text{Hemicellulose} = \%\text{NDF} - \%\text{ADF}$$

$$\%\text{Cellulose} = \%\text{ADF} - (\%\text{AD} - \text{lignin} + \%\text{Ash})$$

NFC was estimated using the equation:[30]

$$\text{NFC}\ (\%\text{of DM}) = 100 - (\text{NDFn} + \text{CP} + \text{FAT} + \text{Ash})$$

where NDFn is 'nitrogen free NDF', it was estimated as:[31]

$$\text{NDFn} = \text{NDF} \times 0.93$$

Packing and scanning by near-infrared spectrometer

Subsamples of the homogenized pulverized samples were packed in circular NIRS ring cups (Part# IH-0386; FOSS North America, Eden Prairie, MN, USA). The cup was overfilled and the excess was scraped away, which resulted in approximately 5-g samples being scanned (approximately 10 mm in depth). We used a FOSS XDS NIR system (FOSS North America) and ISIscan 4.6.11 software (http://www.winisi.com) to scan the samples in reflectance mode. Before scanning any sample, the instrument was warmed-up and satisfactory instrument performance was confirmed via instrument response, photometric repeatability (noise), wavelength accuracy tests and a check cell scan. A sample holder in the instrument held the packed cup and the instrument carried out 32 successive scans covering both visible and NIR regions from 400 to 2498 nm at intervals of 2 nm to give 1049 data points per sample. For the control, the instrument carried out16 scans over an internal standard ceramic disk before and after scanning the samples. Finally, the NIR system referenced the reflectance energy readings for samples to the corresponding readings from the internal standard and recorded the results as the logarithm of the reciprocal of reflectance (log $1/R$, R = reflectance). Vacuum cleaning of the ring cups between packing ensured contamination free scanning of the samples.

Development and validation of NIRS calibration models

For developing and validating the calibration models in the present study, we used the basic methods described in detail earlier by Saha *et al.*[32] and Rushing *et al.*[33] The important differences of the present study compared to the earlier studies[32,33] are adequately described below, allowing an opportunity to follow the protocol for future studies (if needed).

Calibration models development for moisture, DM, TN, CP, ADF, NDF, ash, AD-lignin, cellulose, hemicellulose and NFC utilized absorption of radiation in the whole visible plus near-infrared region (400–2598 nm). Mathematical processing and statistical analysis were performed using the software package WinISI 4.6.11, including the NIR spectra of all152 samples. We used principal components regression analysis in the score program of the WinISI software for scoring and selecting samples for spectral outliers before calibration and validation. The score algorithm ranks spectra according to 'Mahalanobis' distance. The first 'Mahalanobis' distance, called 'global-H' or 'GH' distance, accounts for the difference of a given sample spectrum from the average spectral feature of the sample set as a whole.[34,35] This gives spectral boundaries to eliminate the outlier samples with GH > 3.0. A second 'Mahalanobis' distance, called 'neighboring-H' of 'NH' distance, measures the similarity of a given sample spectrum from that of its nearest neighboring sample and eliminates samples with NH < 0.6. Eliminating samples based on GH and NH distances allows for the development of accurate and robust calibration models.[36] The qualified sample sets remaining after elimination of spectral outliers for moisture, DM, TN, CP, ADF, NDF, ash, AD-lignin, cellulose, hemicellulose and NFC were randomly divided into two subsets: two-thirds for calibration model development and cross-validation and one-third for external validation to test developed model performance. The external validation sample set ensured a true validation of the NIR spectroscopic models for their prediction accuracy based on a set of random samples not included in the calibration sample set set.[37]

We followed the global program in WinISI software as outlined in the WinISI software manual to develop and validate the calibration models. A linear regression method based on a modification of the partial least squares (PLS) algorithm[38] was used for calibration model development. For scatter correction,[39] we used pretreatments with 2,4,4,1 standard normal variate and detrending (SNVD), which represents second derivative treatment of the spectra to optimize calibrations, a gap of 4 (i.e. 4×2 nm = 8 nm, the spacing over which the derivative was calculated) with first smoothing at four data points, and avoidance of second smoothing. This design removed additive baseline and multiplicative signal effects, resulting in a spectrum with zero mean and a variance equal to one. Application of SNVD transformation to raw spectral data reduces spectral differences related to physical characteristics such as particle size and the path length of samples.[39–41] The application of the second-derivative algorithm to the raw spectra (log $1/R$) increases the complexity of spectra and a clear separation between peaks, which overlapped in the raw spectra.[42]

The cross-validation included five steps during calibration development. In each of the five cross-validation steps, the calibration and validation subgroups had 80% and 20% of the samples, respectively. The optimum number of PLS loading factors (called terms) was determined by cross-validation of the MPLS procedure; this was different for different models. Internal cross-validation avoids overfitting of the equations by selecting the minimum number of PLS terms in each model.[40]

Following model development, an additional elimination process removed the compositional outliers from the calibration sample set. The compositional outliers were those samples that had the difference between predicted and laboratory-measured values greater than three times the original standard error of cross-validation (SECV).[43,44] Compositional outliers are considered as the samples with poor quality laboratory-measured values that do not correlate well with the spectral features of the samples.[42–45] The final calibration models were developed following exclusion of the compositional outliers.

A calibration model was considered better when it had a lower the standard error of calibration (SEC) and a higher coefficient of determination for calibration (R^2) that accounts for the proportion of explained variation by the model. The cross-validation ability of a model was considered better when it had a lower overall SECV and a higher '1 − variance ratio statistics (1 – VR)'. The 1 – VR is indeed the coefficient of determination in cross-validation. The following two ratios were utilized to evaluate the quality of the models:[35,46]

(1) RSCD, SD ÷ SECV, the ratio of SECV to deviation (i.e. SD of reference data in calibration set)
(2) RSCIQ, IQ ÷ SECV, the ratio of SECV to IQ distance (i.e. interquartile distance of the reference data in the calibration set)

The spectra of the sample sets were randomly separated out for external validation and were used for prediction of moisture, DM, TN, CP, ADF, NDF, ash, AD-lignin, cellulose, hemicellulose and NFC contents by the calibration models developed. The predicted results were compared with the corresponding

Table 1. Descriptive statistics for the 11 constituents of winter pea used for the development of NIRS calibration models

Constituent	*N*	Range	Mean	SD[a]	Q1[b]	Q2 (Median)	Q3[c]	IQ[d]
Moisture (g kg^{-1})	88	29.90–86.70	50.67	12.53	41.27	48.00	60.23	18.95
Dry-matter (g kg^{-1})	88	913.30–970.10	949.33	12.53	939.78	952.00	958.73	18.95
Total N (g kg^{-1})	97	18.20–48.40	29.11	5.81	25.50	28.50	33.00	7.50
Crude protein (g kg^{-1})	97	113.80–302.50	181.93	36.32	159.40	178.10	206.30	46.90
ADF (g kg^{-1})[e]	95	194.70–484.80	336.35	72.02	281.15	328.60	398.50	117.35
NDF (g kg^{-1})[f]	97	279.20–683.50	448.91	85.33	387.40	429.60	522.20	134.80
AD-lignin (g kg^{-1})	96	27.90–117.80	66.95	18.98	52.85	64.75	79.55	26.70
Ash (g kg^{-1})	96	51.10–159.70	91.07	26.19	69.65	85.40	108.10	38.45
Cellulose (g kg^{-1})	95	152.30–414.40	269.65	59.78	223.85	255.00	317.25	93.40
Hemicellulose (g kg^{-1})	97	5.30–401.50	119.50	55.14	97.90	115.40	136.90	39.00
NFC (g kg^{-1})[g]	96	2.20–397.70	253.83	89.52	203.63	262.30	322.15	118.52

[a] SD, standard deviation of mean.
[b] Q1, first quartile.
[c] Q3, third quartile.
[d] IQ, interquartile distance (IQ = Q3 − Q1).
[e] ADF, acid detergent fiber.
[f] NDF, neutral detergent fiber.
[g] NFC, non-fibrous carbohydrates.

laboratory-measured values as described below. The compositional outlier samples were removed from the validation set if the difference between predicted and laboratory-measured values exceeded three times the original SECV.[40–42] After such an elimination, the remaining samples in the validation set had predicted values with acceptable accuracy. The bias, standard error of prediction (SEP), bias-corrected SEP [SEP_C] and the associated r^2 were then used to evaluate the prediction performance of the models. Additionally, the following two ratios were utilized to evaluate the success of external validation of the models:[35,46]

(1) RPD, SD ÷ SEP, the ratio of performance (SEP) to deviation (i.e. SD of the reference data in the external validation set).
(2) RPIQ, IQ ÷ SEP, the ratio of performance (SEP) to IQ distance (i.e. interquartile distance of the reference data in the external validation set).

The percentage of the variation in the *Y* variable (various quality attributes) that is accounted for by the *X* variable [spectral characteristics, log(1/*R*)] during calibration and external validation are identified by R^2 and r^2, respectively. The RSCD or RSCIQ and the RPD or RPIQ are measures of the coefficient of variation (CV) for the calibration and validation sets, respectively, and represent the factor by which the prediction accuracy increases compared to using the mean composition for the samples; they provide the average errors of prediction during cross-validation and independent validation. Consequently, the RSCD or RSCIQ and RPD or RPIQ relate calibration and validation performance to the range of measurements and are in wide use as a quality indicator of the calibration.[47,48]

RESULTS AND DISCUSSION

Laboratory reference data

The descriptive statistics including range, mean, median, SD, first and third quartiles (Q1 and Q3), and interquartile distance (IQ = Q3 − Q1) for moisture, DM, TN, CP, ADF, NDF, AD-lignin, cellulose and NFC contents of the winter pea samples used as the calibration and validation sets are shown in Tables 1 and 2. The mean value of various constituents tended to be similar between the calibration and validation sets. For example, the mean values were 50.67 *versus* 50.97 g kg^{-1} for moisture, 949.33 *versus* 949.04 g kg^{-1} for DM, 29.11 *versus* 27.68 g kg^{-1} for TN, 181.93 *versus* 173.04 g kg^{-1} for CP, 336.35 *versus* 358.11 g kg^{-1} for ADF, 448.91 *versus* 473.42 g kg^{-1} for NDF, 66.95 *versus* 71.25 g kg^{-1} for AD-lignin, 91.07 *versus* 97.38 g kg^{-1} for ash, 269.65 *versus* 286.87 g kg^{-1} for cellulose, 119.50 *versus* 115.30 g kg^{-1} for hemicellulose, and 253.83 *versus* 231.10 g kg^{-1} for NFC in the calibration and validation sets (Tables 1 and 2), respectively. The median, SD, Q1, Q3 and IQ of the validation sample set were also similar to those of the calibration sample set for most constituents. Most of these observations align with those reported by other studies working with winter peas,[49] alfalfa,[50,51] or several other forage legumes.[51,52] The similarities between the calibration and validation sets suggest that the calibration models developed could reliably be applied to the validation set, without extrapolation from models.[53]

We also obtained 13 alfalfa forage proficiency testing (PT) samples from National Forage Testing Association (NFTA) wet chemistry laboratory certification reports for the years 2014, 2015 and 2016[52] and evaluated the standard error of laboratory reference (SEL) data and its relationship with SD and mean for five forage quality constituents. Table 3 shows the mean, SD, coefficient of variation among samples (CV_{AS}), SEL or repeatability, coefficient of variation among replicates (CV_{AR}) for DM, TN, CP, ADF and NDF derived from the results reported for these 13 alfalfa PT samples analyzed in triplicate by all of the participating laboratories from the USA, Canada and few other countries; this included the University of Georgia Feed and Environmental Water Laboratory where this research was conducted. The SEL was excellent for all five constituents with the CV_{AR} varying from 0.53% to 2.60% only. The mean and SD of this data set for alfalfa PT samples (Table 3) are similar to those of the winter pea sample sets used for calibration (Table 1) and validation (Table 2), suggesting that the SEL values presented in Table 3 could be compared with the SEC and SEP values to reliably evaluate the calibration and validation performance. The dataset for DM content had the lowest diversity with the CV_{AS} of only 0.90%. The TN and CP content had intermediate diversity

Table 2. Descriptive statistics for the 11 constituents of winter pea used in the monitoring (validation) of NIRS calibration models

Constituent	*N*	Minimum	Maximum	Mean	SD[a]	Q1[b]	Q2 (Median)	Q3[c]	IQ[d]	Average GH[e]	Average NH[f]
Moisture (g kg^{-1})	46	32.80	89.30	50.96	11.96	43.13	47.40	55.80	12.67	1.19	0.437
Dry-matter (g kg^{-1})	46	910.70	967.20	949.04	11.96	944.20	952.60	956.87	12.67	1.19	0.437
Total N (g kg^{-1})	49	14.50	39.10	27.68	5.42	24.40	28.60	30.30	5.90	1.17	0.422
Crude protein (g kg^{-1})	49	90.70	244.40	173.04	33.85	152.50	178.80	189.40	36.90	1.17	0.422
ADF (g kg^{-1})	49	190.20	519.20	358.11	80.21	286.90	354.70	418.30	131.40	1.14	0.417
NDF (g kg^{-1})	49	297.60	700.70	473.42	93.16	400.50	466.40	545.80	145.30	1.17	0.422
AD-lignin (g kg^{-1})	49	40.00	122.40	71.25	19.66	57.10	69.20	82.60	25.50	0.943	0.3
Ash (g kg^{-1})	49	54.20	165.60	97.38	29.12	74.40	92.60	113.40	39.00	1.18	0.423
Cellulose (g kg^{-1})	49	144.30	438.50	286.87	69.63	228.70	281.80	351.00	122.30	1.17	0.422
Hemicellulose (g kg^{-1})	49	11.80	195.30	115.30	40.31	94.60	120.10	134.10	39.50	1.17	0.432
NFC (g kg^{-1})	49	23.10	403.30	231.10	95.51	172.80	216.70	315.20	142.40	1.14	0.418

[a] SD, standard deviation of mean.
[b] Q1, first quartile.
[c] Q3, third quartile.
[d] IQ, interquartile distance (IQ = Q3 − Q1).
[e] GH, global-H distance.
[f] NH, neighboring-H distance.

Table 3. Standard error of laboratory reference data and its relationship with SD and mean of the 13 alfalfa forage proficiency testing (PT) samples dispatched by National Forage Testing Association (NFTA) during 2014 to 2016 certification years[a]

Constituent	Number of samples	Mean	SD	CV_{AS}[b]	SEL	CV_{AR}[b]	SD/SEL ratio	Maximum allowed SEL for passing the PT
Dry-matter (g kg^{-1})	13	930.1	8.39	0.90	4.92	0.53	1.71	14.76
Total N (g kg^{-1})	13	28.06	2.40	8.55	0.73	2.60	3.29	2.19
Crude protein (g kg^{-1})	13	175.4	14.99	8.55	4.56	2.60	3.29	13.68
ADF (g kg^{-1})	13	325.3	40.81	12.55	7.69	2.36	5.31	23.07
NDF (g kg^{-1})	13	399.3	46.13	11.55	9.16	2.29	5.04	27.48

SD, standard deviation of mean; CV_{AS}, coefficient of variation among samples (%); SEL, standard error of laboratory reference data; CV_{AR}, coefficient of variation among replicates (%).
[a] Source: NFTA wet chemistry forage testing laboratory certification reports for the years 2014, 2015 and 2016, which provided the mean and Horwitz expected standard deviation (Horwitz, 1982), HSD (used as the SEL here) derived from the triplicate analysis of each sample by all participating laboratories from USA, Canada, and few other countries including the University of Georgia Feed and Environmental Water Laboratory where this research was conducted.
[b] CV = (SD ÷ mean) × 100.

(CV_{AS} of 8.55%). By contrast, NDF and ADF contents had a relatively higher diversity, with CV_{AS} of 12.55 and 11.55, respectively.

Spectroscopic analysis

An average raw NIR reflectance spectrum of the samples is provided in Fig. 1(A). The second derivative was calculated using the log(1/*R*) spectra at gaps of four data points (8 nm) and a smoothing over segments of four data points [2,4,1] with scatter correction (SNVD). The derivative form of an average spectrum is provided in Fig. 1(B).

The main absorption bands in the average raw NIR reflectance spectrum (Fig. 1A) were observed over several wavelengths. The absorption band covering 1436–1464 nm could be a result of several factors, such as C-H combination bands of methylene (-CH_2) group associated with aliphatic and aromatic hydrocarbons, C-H stretching first overtone of starch, N-H stretching first overtone of amide/protein, N-H stretching (symmetric) first overtone of aromatic amines, >C=O stretching third overtone of ketones and aldehydes, and O-H stretching first overtone of water, starch or polymeric alcohol. The band at 1720 nm could be assigned to C-H stretching first overtone of methylene (-CH_2) group associated with aliphatic and aromatic hydrocarbons, and S-H stretching first overtone of thiol group associated with thiols. The band at 1926 nm was related to O-H stretching and HOH deformation combination from molecular water or water molecules in the 3-aminopropyltriethoxysilane-ethanol-water system, O-H stretching and HOH bending combination in polysaccharides, and >C=O stretching first overtone in amides. The band covering 2100–2136 nm could relate to O-H bending or C-O stretching of starch, as well as C-O-O stretching third overtone of starch or cellulose, whereas the band covering 2302–2344 nm was related to C-H bending second overtone of proteins, C-H stretching and CH_2 deformation combination from starch and other polysaccharides, C=O hydrogen bonded to the N-H of the peptide link (termed the α-helix structure) of proteins, C-H stretching (symmetric) first overtone of methylene group associated with aliphatic hydrocarbons, and C-H bending from polysaccharides. The band at 2488 nm was related to C-H stretching and C-C combination of cellulose. The overlaid raw spectra for all 152 samples shown in Fig. 2 reflect the fact that they belong to same population, even though they include multiple genotypes grown at seven different sites in two different growing seasons.

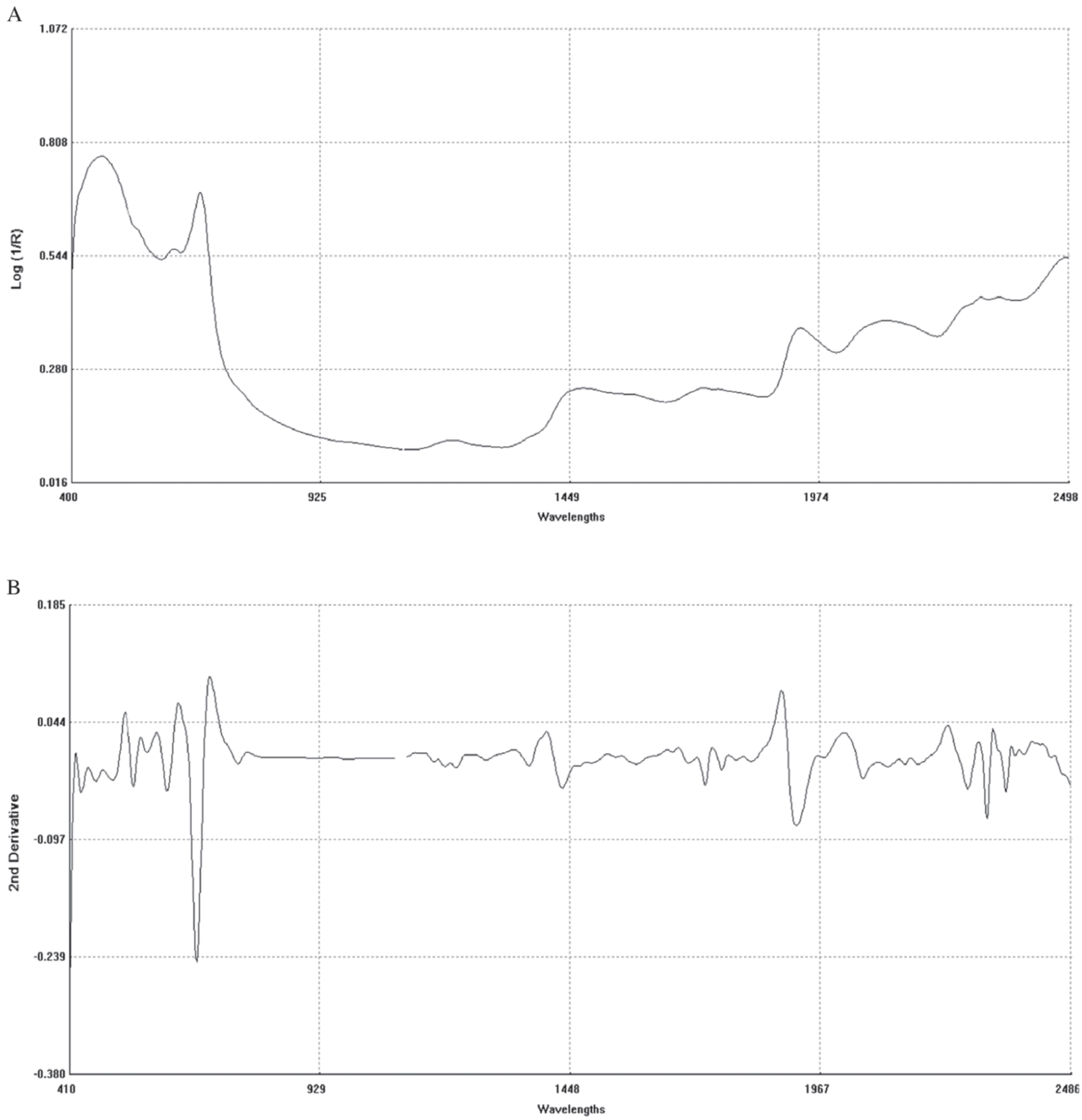

Figure 1. Raw spectrum (A) (log 1/*R*) and second derivative (B) (2,4,4,1 + SNVD) of NIRS average spectrum of a winter pea sample.

The second-derivative spectrum shows a trough corresponding to each peak in the original spectra, removing the overlapping peaks and baseline effects.[53] The second derivative of an average spectrum (Fig. 1B) showed absorption bands at 1398 nm related to C-H bands of methylene (-CH_2) group associated with aliphatic and aromatic hydrocarbons; 1454 nm related to O-H stretching first overtone of starch, C-H bending of methylene (-CH_2) of hydrocarbons and O-H stretching of starch; 1480 nm related to N-H stretching first overtone of $-NH_2C{=}O$; 1526 nm related to N-H stretching first overtone of $-NH_2R$; and 1742 nm related to S-H stretching first overtone of thiol group associated with thiols. The bands at 1780 nm could be a result of C-H stretching first overtone of cellulose and H-O-H deformation and bending of cellulose; those at 1794 nm, a result of O-H bending of water; those at 1962 nm, a result of O-H stretching and O-H bending combination band of starch; those at 2099 nm, a result of O-H bending or C-O stretching third overtone of starch or cellulose; those at 2186 nm, a result of N-H bending second overtone of protein, C-H stretching or C=O bending of protein; those at 2280 and 2348 nm, a result of C-H stretching or -CH_2 deformation-bending of starch; and, those at 2304 nm, a result of C-H bending second overtone of protein.

The NIRS analytical method utilizes the magnitude of absorption/reflection of NIR radiation at a specific wavelength or within a specific region of wavelength by the samples of various natural products. The assignment of main absorption bands in average raw and derivative spectra to various probable functional groups, as described above, was performed in accordance with the relevant previous literature,[54] which showed a good agreement with

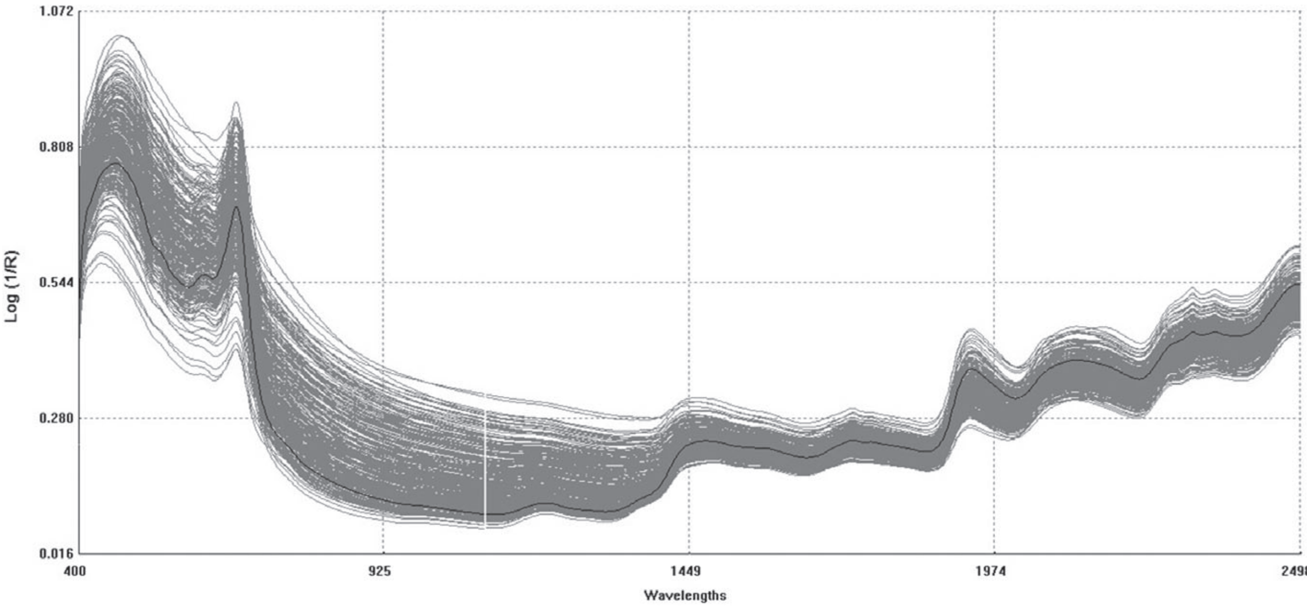

Figure 2. Overlaid raw spectrum (log 1/*R*) of all 152 winter pea samples.

the information for the functional groups in the spectrum given by WinISI software. The wavelength specific absorption/reflection of NIR radiation reflects the presence and abundance of some specific functional groups of various organic compounds in the samples. However, determination of the wavelength(s) or region(s) in the NIR spectrum that carried the most quantitative information about the contents of various natural compounds being analyzed is seldom free from ambiguity, even though the NIRS technique works well in many cases. The available literature also reveals considerable disagreement in the chemical interpretation for absorption/reflection of NIR radiation at a specific wavelength depending on the experimental materials and chemical components that are being considered for NIR analysis.[53–55] Nevertheless, the NIRS technique has been successfully used for determining the contents of various natural compounds in food, feed, biomass, animal wastes and other natural products, even without pinpointing chemical information regarding prominent functional groups related to the NIR spectrum.[32,56–63]

The calibration models

The calibrations and cross-validation statistics of the developed NIRS calibration models for moisture, DM, TN, CP, ADF, NDF, AD-lignin, cellulose ash, and NFC are shown in Table 4. With second-derivative transformation and scatter correction (by 2,4,4,1 SNVD) of raw spectra covering wavelength 400–2498 nm (both visible and NIR range), the MPLS regression procedure produced superior statistics for both calibration (higher R^2 and lower SEC) and cross-validation (higher 1 – VR and lower SECV) than those given by other regression methods tested (actual outcomes not shown). The use of the entire visible-NIR range (400–2498 nm) resulted in much higher R^2 and 1 – VR values and lower values for SEC and SECV than when just the visible (400–1100 nm) or near-infrared (1100–2498 nm) regions were used alone (data not shown). According to Shenk and Westerhaus,[64] optimum wavelengths for NIRS analysis generally relies on empirical calibrations to predict quality constituents for agricultural products because of the broad array of chemical compounds present in the samples, which generally lead to extensively overlapping and perturbed NIR absorption bands.

In addition to R^2 and SEC, 1 – VR and SECV are also valuable statistics for judging any given calibration model because the model may have high R^2 and low SEC but substantially lower 1 – VR (than R^2) and higher SECV (than SEC), suggesting failure of the model in cross-validation and therefore a model with uncertain predictability. In the present study, nine of the 11 calibration models developed (moisture, DM, TN, CP, ADF, NDF, AD-lignin, cellulose and NFC) had a high coefficient of determination (R^2 = 0.88-0.98; 1 – VR = 0.77–0.92) and low standard errors for both calibration and cross-validation, indicating their potential for quantitative prediction ability. However, the calibration models for ash ($R^2 = 0.65$; 1 – VR = 0.46) and hemicellulose ($R^2 = 0.75$; 1 – VR = 0.50) were inferior to the calibration models for the previously mentioned quality constituents but could be good for qualitative screening.

The quality of the models was also evaluated based on another widely used selection criterion, namely RSCD.[55] Besides ash (RSCD = 1.39), all other models had RSCD values greater than 2.0 (moisture, DM, TN, CP, ADF, NDF, AD-lignin, ash, cellulose and NFC at 3.20, 3.20, 3.21, 3.21, 4.47, 5.24, 3.03, 1.39, 6.54, 2.14 and 4.69, respectively), indicating a close relationship between reference values and NIRS predicted values for the samples included in the calibration set.[32,35,46,55]

The RSCIQ is another such criterion reported to be more robust than RSCD because, instead of SD, it is based on interquartile (IQ) distance, which allows for better representation of the spread of the population.[46] The calculated values of RSCIQ were greater than 2.0 for all 11 calibration models developed. Bellon-Maurel *et al.*,[46] proposed that RSCIQ was a judging criterion of the NIRS calibration model, although they did not discuss the interpretation of the situation with a high RSCIQ but a relatively low RSCD, as observed in the present study for ash (RSIQ = 2.14 and RSCD = 1.39). Therefore, it is questionable to accept or reject a calibration model solely based on RSCIQ or RSCD. Due consideration should be given to the other judging criteria (such as R^2, 1 – VR, SEC and SECV).[32]

Table 4. Calibration model development statistics obtained using MPLS and scatter correction (2,4,4,1 SNVD) for the NIRS prediction of the 11 constituents of winter pea

					Calibration		Cross-validation			
Constituent	N^a	Terms[b]	SD	IQ	SEC[c]	R^{2d}	SECV[e]	1 − VR[f]	RSCD[g]	RSCIQ[h]
Moisture (g kg^{-1})	84	3	12.53	18.95	3.91	0.895	4.25	0.874	3.20	4.85
Dry-matter (g kg^{-1})	84	3	12.53	18.95	3.91	0.895	4.25	0.874	3.20	4.85
Total N (g kg^{-1})	97	5	5.81	7.50	1.81	0.903	2.16	0.860	3.21	4.10
Crude protein (g kg^{-1})	97	5	36.32	46.90	11.30	0.903	13.50	0.860	3.21	4.15
ADF (g kg^{-1})	92	7	72.02	117.35	16.10	0.950	23.10	0.898	4.47	7.29
NDF (g kg^{-1})	87	8	80.74	130.80	15.40	0.962	22.40	0.919	5.24	8.49
AD-lignin (g kg^{-1})	88	7	18.95	25.78	6.25	0.875	8.37	0.774	3.03	4.12
Ash (g kg^{-1})	93	2	24.27	35.35	17.40	0.653	18.20	0.460	1.39	2.03
Cellulose (g kg^{-1})	82	10	60.31	96.10	9.22	0.976	16.80	0.921	6.54	10.42
Hemicellulose (g kg^{-1})	80	7	30.97	34.70	14.50	0.753	20.50	0.499	2.14	2.39
NFC (g kg^{-1})	88	8	88.13	116.40	18.80	0.949	27.50	0.890	4.69	6.19

[a] Samples used to develop the model.
[b] Number of PLS loading factors in the regression model MPLS (modified partial least squares).
[c] SEC, standard error of calibration.
[d] R^2, coefficient of determination of calibration.
[e] SECV, standard error of cross-validation.
[f] 1 − Vr, one minus the ratio of unexplained variance divided by variance.
[g] RSCD, SD/SECV, the ratio of standard error of cross-validation to deviation (SD, standard deviation of reference data in calibration set).
[h] RSCIQ, IQ/SECV, the ratio of standard error of cross-validation to interquartile distance (IQ, interquartile distance in reference data in the calibration set).

The laboratory reference values *versus* NIRS predicted values for a selected set of six quality constituents including DM, TN, ADF, NDF, AD-lignin and ash contents for the calibration set are plotted in Fig. 3; such plots remaining quality constituents are not shown for the sake of brevity. The diagonal dashed line in each plot is the 1:1 line. The closeness of the plotted data points to this line represents the closeness between the NIRS predicted values and the corresponding laboratory reference values. These results indicate that the slope regression lines of the measured *versus* predicted values for these six quality constituents are close to 1.00 (0.87–0.95) except ash (0.66). When considered alone, this observation suggests that NIRS/MPLS tends to significantly underestimate ash content. However, other criteria need to be evaluated before reaching such a conclusion. For example, the ash model had 1 – VR noticeably lower than R^2 (0.460 *versus* 0.653, respectively); however, the SECV was not very different from SEC (18.2 *versus* 17.4, respectively), suggesting that the model probably did not lose its performance during cross-validation (Fig. 3). Given this observation, we consider that more emphasis should be given on the external validation statistics to evaluate the model performance.

External validation of the calibration models

The external validation is an essential exercise for judging the predictability of the NIRS calibration models. External validation in the present study had 46 samples for moisture and DM, as well as 49 samples for the remaining nine quality constituents, which were not included in the calibration development process; thus, they provided a true validation opportunity. The r^2, bias, bias (limit) (maximum allowable bias), SEP, SEPc (the bias corrected SEP), SEPc (limit) (the maximum allowable SEPc), slope, RPD (SD/SEP) and RPIQ (IQ/SEP) are important statistics of the external validation exercise (Table 5), allowing a rigorous evaluation of the predictability or reliability of the calibration models. A NIRS calibration model is considered robust and reliable when it demonstrates low bias [lower than the bias (limit)] and SEPc [lower than the SEPc (limit)] with high r^2 and slope close to 1.0 and also RPD and RPIQ values greater than 2.0 in external validation.[32,43,44,56,61,64]

We observed that 86% (42 out of 49) to 100% (49 out of 49) of the samples included in the validation set were successfully predicted with error levels within ± 3 × SECV (compare the number of samples in Tables 2 and 5) for different quality constituents, suggesting that intolerable predictions given by the models were not substantial. It is also possible that the predicted values deviated from the corresponding laboratory measured values beyond the tolerable limit of ± 3 × SECV because of unacceptable errors in the later (laboratory measured values) rather than in the former (predicted values). For all 11 calibration models, the r^2, RPD and RPIQ values obtained during external validation (Table 5) were sufficiently high ($r^2 > 0.8$; RPD and RPIQ > 2.0) to indicate good agreements between the laboratory reference values and the NIRS predicted values. Notably, the r^2 values were greater than 0.9 for six models including, moisture, DM, ADF, NDF, cellulose and NFC. Overall, the validation statistics were better than the calibration and cross-validation statistics. For example, for ash and hemicellulose, the calibration statistics such as R^2, SEC, SECV, RSCD and RSCIQ were 0.653 and 0.753; 17.4 and 14.5; 18.2 and 20.5; 1.39 and 2.14; and 2.03 and 2.39, respectively (Table 4). By contrast, the validation statistics such as r^2 (comparable to R^2), SEP (comparable to SEC and SECV), RPD (comparable to RSCD) and RPIQ (comparable to RSCIQ) for ash and hemicellulose were 0.832 and 0.840; 11.0 and 14.1; 2.42 and 3.47; and 3.01 and 2.17, respectively (Table 5). The slopes for the 11 models ranged from 0.94 to 1.10, with eight of them ranging from 0.95 to 1.05, suggesting that all 11 models validated well yielding reliable quantitative predictions. Therefore, they are suitable for routine analysis of all 11 quality constituents.

In the absence of winter pea-specific calibration models, many laboratories use the mixed hay calibration model released by the NIRS Feed and Forage Testing Consortium. Therefore, it would

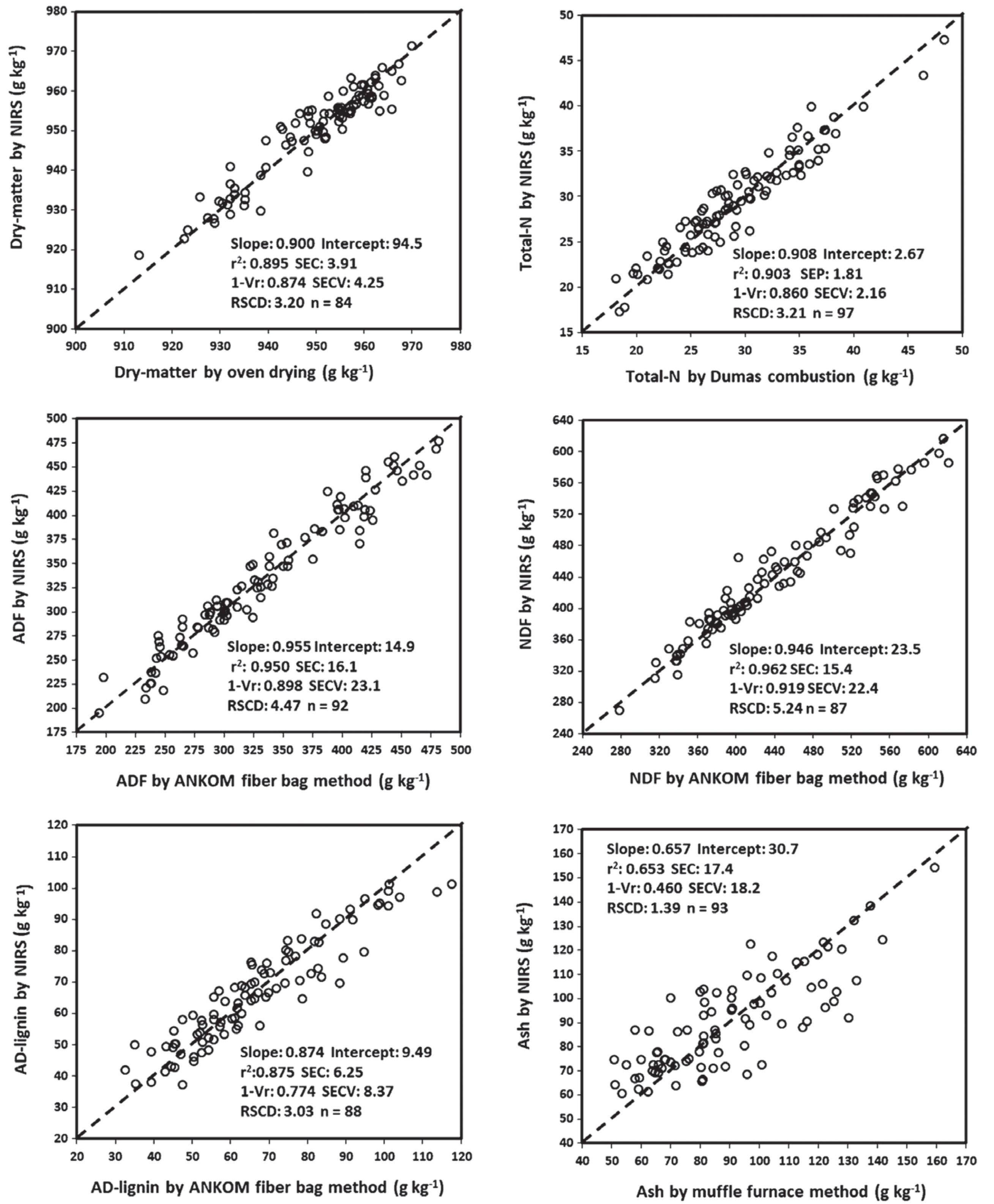

Figure 3. Scatter plots of NIRS predicted values *versus* laboratory reference values for calibration sets of some selected quality constituents of winter pea.

be appropriate to compare the errors of the calibration models developed in the present study with those of the consortium's mixed hay calibration model. The SEP values of the calibration models developed in the present study *versus* the corresponding SECV values of the mixed hay calibration model released by consortium (designated as '16mh50-2.eqa') are 3.55 *versus* 3.11 g kg^{-1} for both moisture and DM, 2.02 *versus* 1.41 g kg^{-1} for TN, 12.63 *versus* 8.80 g kg^{-1} for CP, 22.83 *versus* 18.86 g kg^{-1} for ADF, 24.19 *versus* 23.30 g kg^{-1} for NDF, 7.56 *versus* 12.51 g kg^{-1} for AD-lignin and 11.12 *versus* 10.79 g kg^{-1} for ash. Thus, the

Table 5. Monitoring (external validation) statistics for the NIR spectroscopic prediction equations developed with 2,4,4,1 math treatment for 11 constituents of winter pea

Constituent	N^a	SD^b	IQ^c	$Bias^d$	Bias (limit)[e]	SEP^f	$SEPc^g$	SEPc (limit)[h]	r^{2i}	Slope[j]	Intercept[j]	RPD^k	$RPIQ^l$
Moisture (g kg^{-1})	45	12.00	12.10	0.380	2.55	3.55	3.57	5.53	0.913	0.962	2.21	3.38	3.41
Dry-matter (g kg^{-1})	45	12.00	12.10	−0.380	2.55	3.55	3.57	5.53	0.913	0.962	48.59	3.38	3.41
Total N (g kg^{-1})	49	5.42	5.90	−0.250	1.30	2.02	2.03	2.81	0.876	0.994	0.410	2.68	2.92
Crude protein (g kg^{-1})	49	33.85	36.90	−1.560	8.11	12.63	12.66	17.56	0.876	0.994	2.70	2.68	2.92
ADF (g kg^{-1})	48	81.00	132.8	3.41	13.84	22.83	22.81	29.98	0.921	1.017	30.33	3.55	5.82
NDF (g kg^{-1})	49	93.16	145.3	9.77	13.46	24.19	22.36	29.16	0.948	1.080	48.34	3.85	6.01
AD-lignin (g kg^{-1})	42	17.42	23.80	−0.830	5.02	7.56	7.60	10.88	0.825	0.940	4.97	2.30	3.15
Ash (g kg^{-1})	42	26.62	33.08	−0.500	10.90	11.00	11.12	23.	0.832	1.096	23.09	2.42	3.01
Cellulose (g kg^{-1})	49	69.63	122.3	7.60	13.94	22.15	21.02	30.20	0.910	1.032	26.42	3.14	5.52
Hemicellulose (g kg^{-1})	42	34.71	30.55	−2.76	12.31	14.08	13.98	26.68	0.840	0.952	16.68	3.47	2.17
NFC (g kg^{-1})	47	96.40	137.6	0.180	16.52	27.47	27.77	35.80	0.918	1.029	25.13	3.51	5.01

[a] Samples (independent) used to monitor the model.
[b] SD, standard deviation of mean.
[c] IQ, interquartile distance (IQ, interquartile distance in reference data).
[d] Bias, average difference between reference and NIRS values.
[e] Bias(limit) maximum allowable value of bias.
[f] SEP, standard error of prediction.
[g] SEPc, the bias-corrected standard error of prediction.
[h] SEPc(limit), the maximum allowable value of SEPc.
[i] r^2, coefficient of determination in external validation.
[j] Slope and intercept, the steepness and intercept of a straight line curve for the plot between the NIR predicted values *versus* reference values.
[k] RPD, SD/SEP, the ratio of performance to deviation (SEP to the SD of the reference data in the external validation set).
[l] RPIQ, IQ/SEP, the ratio of performance to interquartile distance (SEP to the IQ of the reference data in the external validation set).

error levels in the calibration models of the present study and those of '16mh50-2.eqa' are mostly comparable. The observed difference in error levels for a few constituents between the two calibration reports could be attributed to the differences in the spread of the data sets used in two cases. Saha *et al.*[32] reported NIRS calibration models for a combined soybean and sunflower biomass population with SEP values of 3.10 g kg^{-1} for moisture and DM, 1.20 g kg^{-1} for TN, 7.47 g kg^{-1} for CP, 14.35 g kg^{-1} for ADF, 10.86 g kg^{-1} for NDF, 5.13 g kg^{-1} for AD-lignin and 8.12 g kg^{-1} for ash. With wild rye (*Elymus glabriflrous*), Rushing *et al.*[33] reported NIRS calibration models with SEP values of 5.23 g kg^{-1} for moisture and DM, 6.65 g kg^{-1} for CP, 23.95 g kg^{-1} for ADF and 24.51 g kg^{-1} for NDF.

The plots of NIRS predicted values *versus* laboratory reference values for a selected set of six quality constituents including DM, TN, ADF, NDF, AD-lignin and ash contents for the validation set are shown in Fig. 4. All six scatter plots show the significant relationship between NIRS and the reference laboratory method.

Another very important question is whether the winter pea biomass samples could be accurately analyzed by the mixed hay calibration model '16mh50-2.eqa' released by the NIRS Feed and Forage Testing Consortium for analyzing various legumes; in other words, whether a winter pea-specific calibration model is essential given the fact that the mixed hay calibration model '16mh50-2.eqa' released by the NIRS Feed and Forage Testing Consortium is in use for analysis of various legumes. To address this question, the same external validation set of winter pea samples (included in Table 5) were predicted by the '16mh50-2.eqa' and the predicted results were statistically validated against the corresponding laboratory reference values. The validation statistics such as r^2, bias, bias (limit) (maximum allowable bias), SEP, SEPc (the bias corrected SEP), SEPc (limit) (the maximum allowable SEPc), slope, RPD (SD/SEP) and RPIQ (IQ/SEP) are presented in Table 6. Generally, the various validation statistics in Table 6 are clearly inferior to the corresponding ones in Table 5. Particularly, actual prediction 'Bias' is much higher than the 'Bias(limit)' for all six constituents. Likewise, the SEPc values were invariably much higher than SEPc(limit) values. Such results suggest that the accuracy of analyzing winter pea sample by the mixed hay calibration model '16mh50-2.eqa' is not acceptable. This outcome was quite expected given the fact that relative to the spectral population used for developing the '16mh50-2.eqa' calibration model, the ranges of average GH and NH distances for the winter pea validation sample set were high and ranged from 5.69 to 5.92 and from 4.05 to 4.43, respectively, for different constituents, suggesting that analysis of winter pea samples by '16mh50-2.eqa' calibration model is unacceptable for winter pea analysis because it would be an 'extrapolation' rather than a 'prediction'. Thus, it is clear that the calibration model specifically for winter pea developed and validated in the present study was necessary for accurate analysis of this crop.

Final calibration models combining calibration and validation set

The calibration and validation sample sets were combined for final calibration model development. The statistics for the final calibration models are presented in Table 7. Nine of the 11 final models yielded a high coefficient of determination ($R^2 = 0.89-0.95$; $1 - VR = 0.87-0.94$) and a low standard error in both calibration and cross-validation. The calibration models for ash ($R^2 = 0.66$; $1 - VR = 0.62$) and hemicellulose ($R^2 = 0.81$; $1 - VR = 0.64$) not only had a lower coefficient of determination, but also appeared appropriate for qualitative screening. Figure 5 shows the plots of laboratory reference values *versus* NIRS predicted values for a selected set of six quality constituents including DM, TN, ADF, NDF, AD-lignin and ash contents for the final calibration sample set, all of which show a significant positive relationship between the dependent and independent variables.

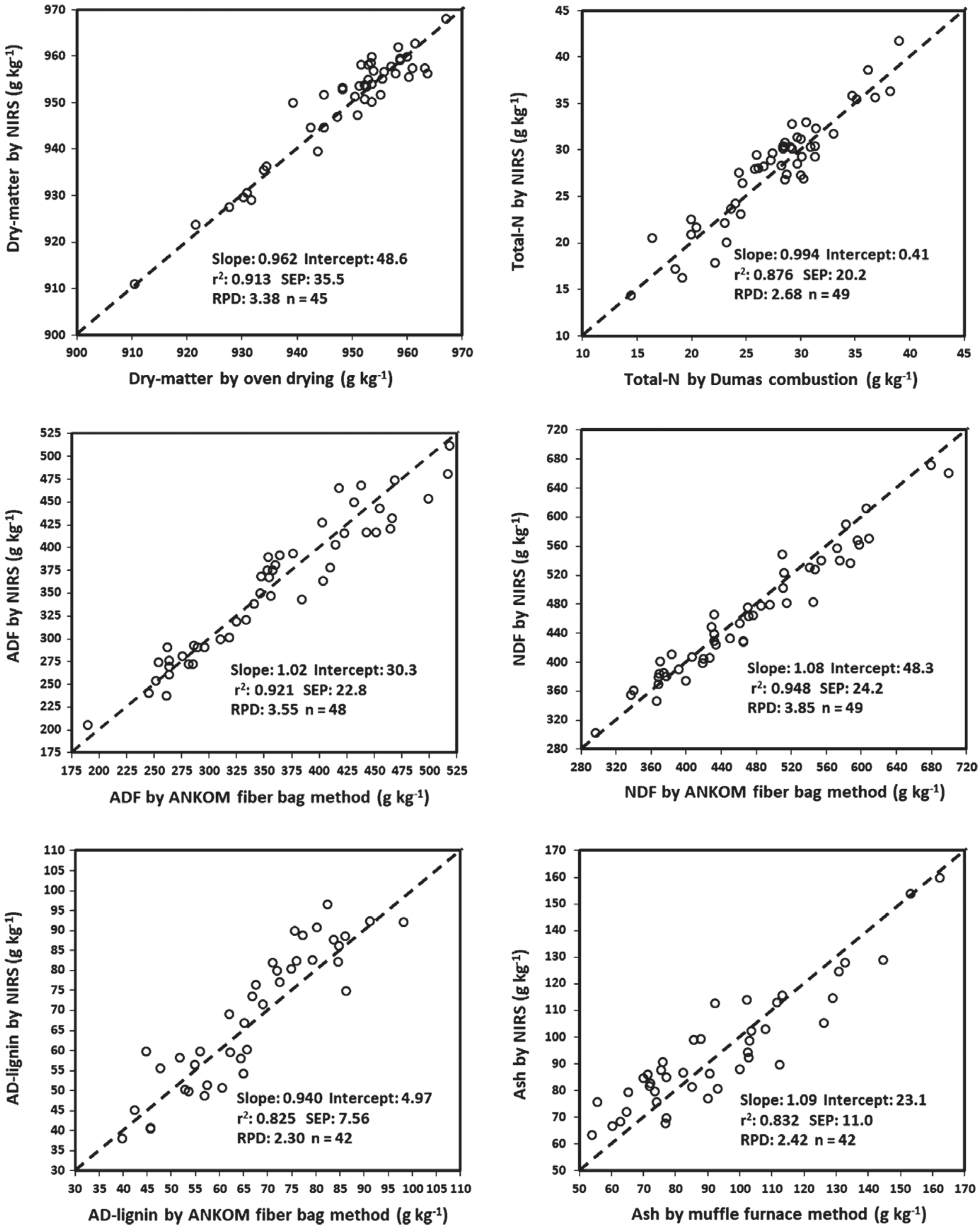

Figure 4. Scatter plots of NIRS predicted values *versus* laboratory reference values for external validation sets of some selected quality constituents of winter pea.

Table 6. Monitoring (external validation) statistics for the NIR spectroscopic 'general mixed legume prediction equation: 16mh50-2-eqa' obtained from 'NIRS forage and feed testing consortium (https://www.nirsconsortium.org/)' for six primary constituents of winter pea samples

Constituent	*N*[a]	SD[b]	IQ[c]	Bias[d]	Bias (limit)[e]	SEP[f]	SEPc[g]	SEPc (limit)[h]	r2[i]	Slope[j]	Intercept[j]	RPD[k]	RPIQ[l]
Dry-matter (g kg^{-1})	45	12.00	12.10	19.50	18.70	7.200	7.010	4.040	0.774	1.075	73.33	1.67	1.68
Crude protein (g kg^{-1})	49	33.85	36.90	11.75	5.280	24.10	21.29	11.44	0.606	0.573	62.20	1.40	1.53
ADF (g kg^{-1})	48	81.00	132.88	−21.38	11.32	43.70	38.53	24.52	0.770	0.787	97.82	1.85	3.04
NDF (g kg^{-1})	49	93.16	145.30	−58.72	13.98	66.77	32.13	30.29	0.883	0.922	95.62	1.40	2.18
AD-lignin (g kg^{-1})[j]	42	17.42	23.80	−9.130	6.470	16.14	13.45	14.03	0.552	0.658	33.49	1.08	1.47
Ash (g kg^{-1})	42	26.62	33.08	36.62	1.300	47.40	30.45	28.10	0.088	0.214	39.94	0.56	0.700

[a] Samples (independent) used to monitor the model.
[b] SD, standard deviation of mean.
[c] IQ, interquartile distance (IQ, interquartile distance in reference data).
[d] Bias, average difference between reference and NIRS values.
[e] Bias(limit) maximum allowable value of bias.
[f] SEP, standard error of prediction.
[g] SEPc, the bias-corrected standard error of prediction.
[h] SEPc(limit), the maximum allowable value of SEPc.
[i] r^2, coefficient of determination in external validation.
[j] Slope and intercept, the steepness and intercept of a straight line curve for the plot between the NIR predicted values *versus* reference values.
[k] RPD, SD/SEP, the ratio of performance to deviation (SEP to the SD of the reference data in the external validation set).
[l] RPIQ, IQ/SEP, the ratio of performance to interquartile distance (SEP to the IQ of the reference data in the external validation set).
[m] AD-lignin means acid detergent lignin.
The average GH distance of the validation set NIRS spectra relative to the spectral population of the 'mixed legume prediction equation: 16mh50-2-eqa' obtained from 'NIRS forage and feed testing consortium' ranged from 5.69 to 5.92 for various parameters. The range of average NH distance for the same context was from 4.03 to 4.45.

Table 7. Statistics of final calibration model development obtained using MPLS and scatter correction (2,4,4,1 SNVD) for the NIRS prediction of the 11 constituents of winter pea

					Calibration		Cross-validation			
Constituent	*N*[a]	Terms[b]	SD	IQ	SEC[c]	R2[d]	SECV[e]	1 − VR[f]	RSCD[g]	RSCIQ[h]
Moisture (g kg^{-1})	128	4	12.38	14.73	3.30	0.929	3.66	0.913	3.76	4.47
Dry-matter (g kg^{-1})	128	4	12.38	14.73	3.30	0.929	3.66	0.913	3.76	4.47
Total N (g kg^{-1})	145	6	5.72	7.20	1.78	0.904	2.05	0.871	3.22	4.06
Crude protein (g kg^{-1})	145	6	35.77	45.00	11.11	0.904	12.85	0.871	3.22	4.05
ADF (g kg^{-1})	142	7	75.74	124.2	16.80	0.951	20.85	0.924	4.51	7.39
NDF (g kg^{-1})	141	5	86.2	131.7	19.71	0.948	21.86	0.936	4.37	6.68
AD-lignin (g kg^{-1})	145	7	19.26	26.80	6.34	0.892	7.54	0.847	3.04	4.23
Ash (g kg^{-1})	144	3	26.92	40.03	15.58	0.665	16.49	0.625	1.73	2.57
Cellulose (g kg^{-1})	142	9	63.51	93.48	14.89	0.945	21.59	0.884	4.26	6.28
Hemicellulose (g kg^{-1})	129	9	29.98	35.20	13.09	0.809	18.07	0.637	2.29	2.69
NFC (g kg^{-1})	141	6	91.85	129.5	25.39	0.924	28.99	0.900	3.62	5.10

[a] Samples used to develop the model.
[b] Number of PLS loading factors in the regression model MPLS (modified partial least-squares).
[c] SEC, standard error of calibration.
[d] R^2, coefficient of determination of calibration.
[e] SECV, standard error of cross-validation.
[f] 1 − Vr, one minus the ratio of unexplained variance divided by variance.
[g] RSCD, SD/SECV, the ratio of standard error of cross-validation to deviation (SD, standard deviation of reference data in calibration set).
[h] RSCIQ, IQ/SECV, the ratio of standard error of cross-validation to interquartile distance (IQ, interquartile distance in reference data in the calibration set).

Despite the abundant advantages of NIRS, its predictions become inaccurate when improperly used or abused by predicting samples with spectral characteristics dissimilar from the original calibration sample set. Such differences in spectral feature can be quantified by GH and NH distances described in the Materials and methods. These differences can result from a variety of factors, including differences in varieties/cultivars, cultivation area, influences of genetic and environmental variation, and many others. However, these problems can be minimized by updating, expanding and improving the initially developed and validated calibrations (such as the ones presented here). This can be achieved by including future samples from different environments and varieties/cultivars, as well as by covering a wider range for the predicted constituents, thereby furthering the robustness of the current calibration models. This should be a routine practice for any NIRS laboratory.

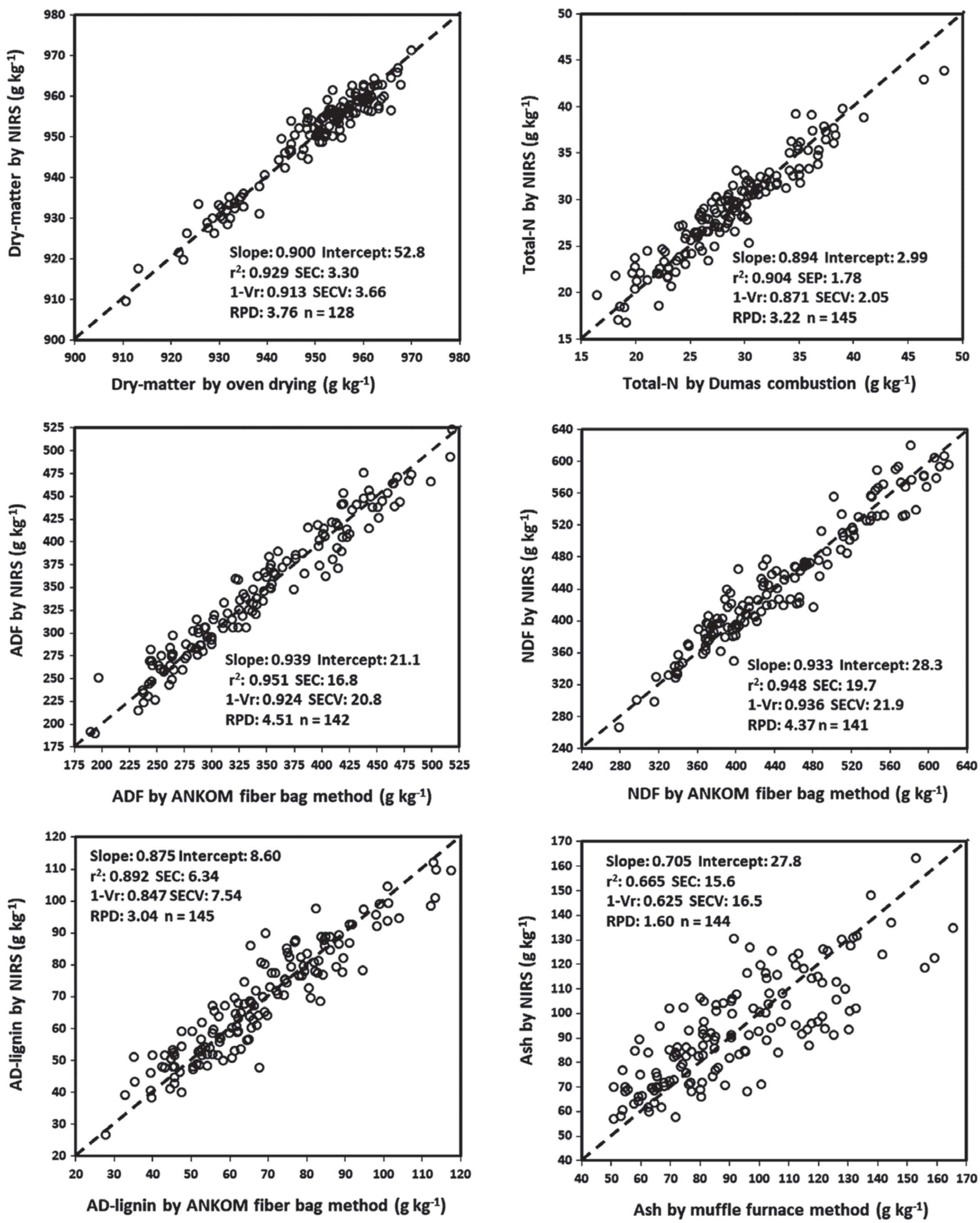

Figure 5. Scatter plots of NIRS predicted values *versus* laboratory reference values for final calibration sets (combining all samples) of some selected quality constituents of winter pea.

CONCLUSIONS

The chemometric techniques applied on this winter pea spectral population established a valid quantitative relationship between the visible-NIR light absorption characteristics and laboratory measured values of 11 forage and cover crop quality constituents of winter pea, thereby developing acceptable NIRS calibration models for predicting these quality constituents in this species. All 11 developed calibration models had prediction abilities with acceptable accuracy; nine of them appeared suitable for quantitative prediction and the other two at least for qualitative screening, if not for quantitative prediction. These models can currently be reliably applied in the routine analysis of these properties of winter pea. These models could also find application in routine analysis in wider geographic territories with the expansion of model development using samples from other growing regions and seasons. Once the samples are ground as described in the present study, this non-destructive NIRS method is a rapid and cost-effective alternative to using wet-chemistry analyses. The use of the NIRS method could simplify the analysis of the constituents of interest, replacing tedious and expensive wet-chemistry based laboratory methods.

The developed and validated NIRS calibration models presented for the 11 constituents in the present study will serve both investigating researchers and the ultimate end-users producing winter pea for forage or cover crops. This method provides an alternative to time-consuming, costly laboratory methods and allows for rapid and high throughput analysis Additional benefits of NIRS predicted quality constituents include a reduced cost and a lack of chemical use in the sample analyses. To our knowledge, this is the first report of original research aiming to develop NIRS calibration models for analyzing winter pea for its forage and cover crop quality profile in the USA. Although the calibration models developed in the present study included winter pea samples from the southeastern USA only, these base models provide foundational information and could be expanded to include winter pea produced in other regions within and outside the USA for broader use.

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